

*Compact dual-band L1/L5 multipath mitigation GNSS device based on U-blox **DAN-F10N**. Embedded L1/L5 GNSS antenna. Onboard **RM3100** magnetometer and **BMP581** barometric sensor.*

*Datasheet - In Production*



## Features

- 5V - 60mA power supply
- LEDs status: Power/Timepulse
- Timepulse output
- I<sup>2</sup>C/UART digital interfaces
- Onboard Bosch BMP581 barometer
- Onboard PNI RM3100 compass
- Onboard L1/L5 GNSS antenna

## Applications

- Drones
- Ground vehicles
- Precise navigation

## Description

The Siderion **Polaris F10N** is an affordable, compact standard precision dual-band patch antenna module based on the u-blox F10 L1/L5 GNSS device. It provides robust meter-level positioning accuracy in challenging environments.

Using u-blox **DAN-F10N** proprietary dual-band multipath mitigation technology, the Polaris F10N utilises the best signals from the L1/L5 bands to achieve better accuracy in urban environments.

The onboard **BMP581** absolute barometric pressure sensor and the professional-grade **RM3100** compass enhance the overall performance of the Polaris F10N.

The JST-GH connectors make them perfect to be connected to a Pixhawk or any autopilots.

The Polaris F10N operates across a temperature range from -40°C to +70°C.

Table 1. Device summary

| Order ref code        | Temperature [°C] | Product size [mm]  | Weight [g] |
|-----------------------|------------------|--------------------|------------|
| SID-GNSS-POLARIS_F10N | -40 to +70       | 38.0 x 38.0 x 15.0 | 30         |

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# 1. Block diagram and pin description

## 1.1 Block diagram

Figure 1. Polaris F10N block diagram & connectivity

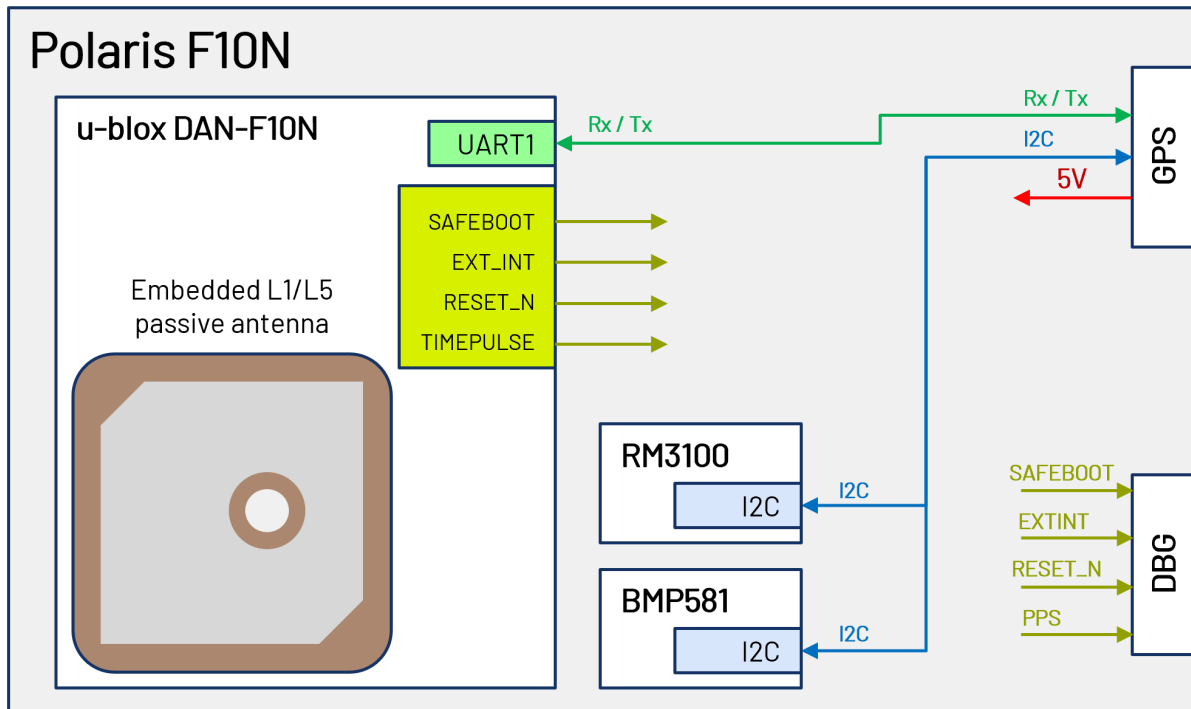
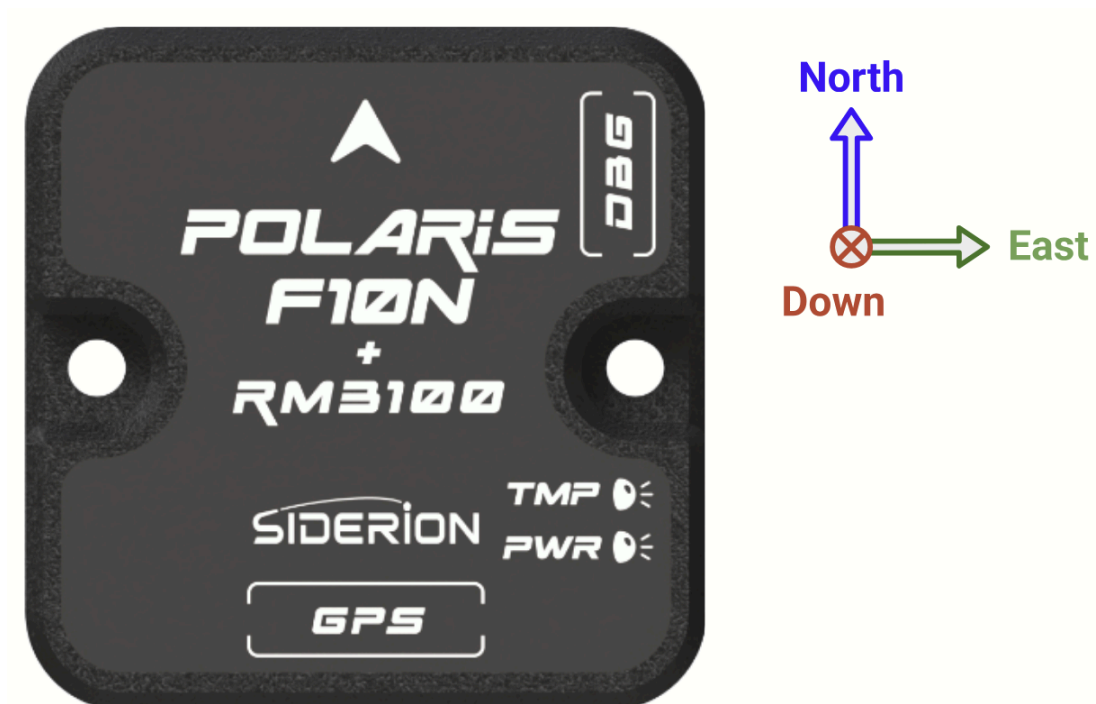


Figure 2. Polaris F10N North-East-Down (NED) orientation (top view)



## 1.2 LED description

Table 2. Polaris F10N LED sequence status

| LED name  | Color | Light sequence                                                                    | Comment                                        |
|-----------|-------|-----------------------------------------------------------------------------------|------------------------------------------------|
| Power     | Blue  |  | Solid blue LED when powered ON                 |
| Timepulse | Green |  | No Timepulse                                   |
|           |       |  | Timepulse LED is blinking at the selected rate |

## 1.3 Pin description

Figure 2. Polaris F10N pinout (top view & side views)



Table 3. Pinout configuration

|     | Pin | Name      | Type | Function                 |
|-----|-----|-----------|------|--------------------------|
| GPS | A1  | 5V IN     | P    | 5V input                 |
|     | A2  | UART1 RXD | I    | UART1 receive            |
|     | A3  | UART1 TXD | O    | UART1 transmit           |
|     | A4  | I2C SCL   | I    | I2C clock                |
|     | A5  | I2C SDA   | I/O  | I2C data                 |
|     | A6  | GND       | P    | Ground reference         |
| DBG | B1  | TIMEPULSE | O    | Timepulse output         |
|     | B2  | nRESET    | I    | Reset input              |
|     | B3  | EXTINT    | I    | External Interrupt input |
|     | B4  | nSAFEBOOT | I    | Safe Boot input          |

P : Power / I : Input / O : Output

## 2. Absolute maximum ratings

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 4. Polaris F10N absolute maximum ratings

| Symbol | Parameter               | Maximum value   | Units |
|--------|-------------------------|-----------------|-------|
| Vusb   | USB supply voltage      | -0.3 to +6      | V     |
| Vdd    | Internal supply voltage | -0.5 to +3.6    | V     |
| Vdd_IO | I/O pins supply voltage | -0.5 to Vdd+0.1 | V     |
| Icc_RF | RF output current       | 50              | mA    |
| Prfin  | Input power at RF_IN    | 0               | dBm   |
| W      | Product weight          | 30              | g     |
| TOP    | Operating temperature   | -40 to +70      | °C    |
| TSTG   | Storage temperature     | -40 to +80      | °C    |



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.

## 3. Specifications

### 3.1 Ublox DAN-F10N specifications

Table 5. DAN-F10N mechanical and electrical specifications

| Symbol | Parameter                        | Test conditions | Min.    | Typ. | Max.    | Units |
|--------|----------------------------------|-----------------|---------|------|---------|-------|
| Vusb   | USB supply voltage               |                 | 4.5     | 5.0  | 5.5     | V     |
| Vdd    | Internal supply voltage          |                 |         | 3.3  |         | V     |
| Vdd_IO | Supply voltage for I/O           |                 |         | 3.3  |         | V     |
| Idd    | Current consumption              |                 |         |      | 50      | mA    |
| Vil    | IO pin low level input voltage   |                 | 0       |      | 0.8     | V     |
| Vih    | IO pin high level input voltage  |                 | 2       |      | Vdd+0.3 | V     |
| Vol    | IO pin low level output voltage  | Iol = 1mA       |         |      | 0.4     | V     |
| Voh    | IO pin high level output voltage | Ioh = 1mA       | Vdd-0.4 |      |         | V     |
| Top    | Operating temperature            |                 | -40     |      | +85     | °C    |

Table 6. DAN-F10N general performance

| Parameter              | Specifications                   | Value                       |
|------------------------|----------------------------------|-----------------------------|
| Receiver type          | Multi-band GNSS high precision   |                             |
| Accuracy of Timepulse  | RMS<br>99%                       | 30 ns<br>60 ns              |
| Frequency of Timepulse |                                  | 0.25 Hz to 10 MHz           |
| Operational limits     | Dynamics<br>Altitude<br>Velocity | < 4g<br>80,000 m<br>500 m/s |
| Velocity accuracy      |                                  | 0.03 m/s                    |

Table 7. DAN-F10N performance in different GNSS mode

| GNSS                     | Parameter       | GPS+GAL+BDS | GPS+BDS   | GPS+GAL   | GPS       |
|--------------------------|-----------------|-------------|-----------|-----------|-----------|
| Acquisition              | Cold start      | 28 s        | 28 s      | 27 s      | 27 s      |
|                          | Hot start       | 2 s         | 2 s       | 2 s       | 2 s       |
|                          | Aided start     | 2 s         | 2 s       | 2 s       | 2 s       |
| Update rate              |                 | 10 Hz       | 10 Hz     | 10 Hz     | 20 Hz     |
| Horizontal pos. accuracy | PVT             | 1.0 m CEP   | 1.0 m CEP | 1.0 m CEP | 1.5 m CEP |
| Sensitivity              | Tracking & Nav. | -164 dBm    | -164 dBm  | -164 dBm  | -164 dBm  |
|                          | Reacquisition   | -156 dBm    | -156 dBm  | -156 dBm  | -156 dBm  |
|                          | Cold start      | -145 dBm    | -145 dBm  | -145 dBm  | -145 dBm  |
|                          | Hot start       | -156 dBm    | -156 dBm  | -156 dBm  | -156 dBm  |

Table 8. DAN-F10N supported GNSS constellations and signals

| System  | Signal             | Frequency                    |
|---------|--------------------|------------------------------|
| GPS     | L1C/A<br>L5        | 1575.420 MHz<br>1176.450 MHz |
| Galileo | E1-B/C<br>E5a      | 1575.420 MHz<br>1176.450 MHz |
| BeiDou  | B1C<br>B2a         | 1575.420 MHz<br>1176.450 MHz |
| QZSS    | L1C/A, L1C/B<br>L5 | 1575.420 MHz<br>1176.450 MHz |
| NavIC   | L5-SPS             | 1176.450 MHz                 |

## 3.2 PNI RM3100 specifications

Table 9. RM3100 electrical specifications

| Symbol     | Parameter                                                    | Test conditions | Min.     | Typ. | Max.     | Units |
|------------|--------------------------------------------------------------|-----------------|----------|------|----------|-------|
| Vdd        | Internal supply voltage                                      |                 | 2.0      | 3.0  | 3.6      | V     |
| Vdd_IO     | Supply voltage for I/O                                       |                 | 2.0      | 3.0  | 3.6      | V     |
| Vdd_Ripple | Vdd ripple voltage                                           |                 |          |      | 0.05     | Vpp   |
| Idd        | Current consumption                                          |                 |          |      | 1        | mA    |
| Vil        | IO pin low level input voltage                               |                 | 0        |      | 0.3* Vdd | V     |
| Vih        | IO pin high level input voltage                              |                 | 0.7* Vdd |      | Vdd      | V     |
| Iidle      | Idle Mode Current                                            |                 |          |      | 1        | μA    |
| Ileak      | Leakage Current @ DVDD pin<br>(AVDD=AVSS=DVSS=0V, DVDD=3.6V) |                 |          |      | 100      | nA    |
| Top        | Operating temperature                                        |                 | -20      |      | +85      | °C    |
| I2Cadd     | 7bit I2C address                                             | 0b0100000       |          |      |          |       |

Table 10. RM3100 Geomagnetic Sensor Performance

| Parameter                                    | Cycle counts |     |     | Units  |
|----------------------------------------------|--------------|-----|-----|--------|
|                                              | 50           | 100 | 200 |        |
| Field Measurement Range                      | -800 to +800 |     |     | μT     |
| Gain                                         | 20           | 38  | 75  | LSB/μT |
| Sensitivity                                  | 50           | 26  | 13  | nT     |
| Noise                                        | 30           | 20  | 15  | nT     |
| Noise Density @ Max. Single-Axis Sample Rate | 1.2          |     |     | nT/√Hz |
| Repeatability over ±200 μT                   | 15           | 8   | 8   | nT     |
| Hysteresis over ±200 μT                      | 15           |     |     | nT     |
| Linearity over ±200 μT                       | 0.5          |     |     | %      |

|                                                                               |      |     |     |     |
|-------------------------------------------------------------------------------|------|-----|-----|-----|
| Maximum Single-Axis Sample Rate<br>(divide by 3 for max. 3-axis sample rate)  | 1600 | 850 | 440 | Hz  |
| Single-Axis Average Current @ 24 Hz Sample Rate (equivalent to 3-axis @ 8 Hz) | 70   | 135 | 260 | μA  |
| Bias Resistor, Rb                                                             | 121  |     |     | Ω   |
| Ext. Timing Resistor for Clock, REXT                                          | 33   |     |     | KΩ  |
| Circuit Oscillation Frequency                                                 | 180  |     |     | kHz |

### 3.3 Bosch Sensortech BMP581 specifications

Table 11. BM581 electrical specifications

| Symbol   | Parameter                           | Test conditions                | Min. | Typ. | Max.        | Units    |
|----------|-------------------------------------|--------------------------------|------|------|-------------|----------|
| Type     | Absolute barometric pressure sensor |                                |      |      |             |          |
| Pres     | Pressure Range                      |                                | 30   |      | 125         | kPa      |
| Vdd      | Supply Voltage (VDD)                |                                | 1.65 |      | 3.6         | V        |
| Idd      | Current Consumption                 | Peak<br>Deep standby           |      |      | 260<br>0.55 | μA       |
| Abac     | Absolute Accuracy                   | -5°C to 65°C                   |      | ±30  |             | Pa       |
| Reac     | Relative Accuracy                   | 700 to 1100 hPa                |      | ±6   |             | Pa       |
| Reac     | Relative Accuracy                   | 900 to 1100 hPa                |      | ±0.4 |             | Pa       |
| Noise    | Noise (Lowest Bandwidth)            |                                |      | 0.08 |             | Pa       |
| Temp_off | Temperature Coefficient Offset      |                                |      | ±0.5 |             | Pa       |
| ltd      | Long-term Drift (12 months)         |                                |      | ±0.1 |             | Pa       |
| Sd       | Solder Drift                        |                                |      | ±30  |             | Pa       |
| ODR      | Output Data Rate                    | continuous mode<br>normal mode |      |      | 480<br>240  | Hz<br>Hz |
| Int      | Interfaces                          | I <sup>2</sup> C               |      |      | 1           | MHz      |
| Temp     | Temperature Range                   |                                | -40  |      | 85          | °C       |
| I2Cadd   | 7bit I2C address                    | 0b1000110                      |      |      |             |          |

## 4. Applications

### 4.1 Standalone configuration

## 5. Communication interfaces

There are several communications interfaces including UART and I2C. UART is used to communicate with the DAN-F10N whereas I2C is used to communicate with both embedded sensors: RM3100 & BM581. All the inputs have internal pull-up resistors in normal operation and can be left open if not used. All the PIOs are supplied by VCC, therefore all the voltage levels of the PIO pins are related to Vdd supply voltage.

### 5.1 UART Interface

There is one UART interface: UART. UART operates up to and including a speed of 1 Mbaud. No hardware flow control on UART is supported.

Figure 4. UART chronogram specifications

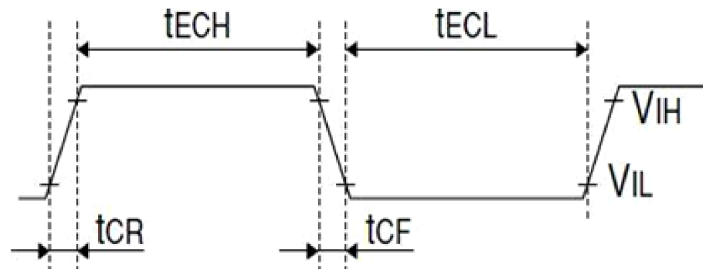


Table 12. DAN-F10N serial UART timing specifications

| Symbol | Parameter                          | Min.    | Max.    | Units |
|--------|------------------------------------|---------|---------|-------|
| Vil    | LOW-LEVEL input voltage            | 0       | 0.2xVdd | V     |
| Vih    | HIGH-LEVEL input voltage           | 0.7xVdd | Vdd+0.3 | V     |
| tECH   | HIGH period of external data input | 0       | 0.4     | μs    |
| tECL   | LOW period of external data input  | TBA     | TBA     | μs    |
| Ru     | Baudrate                           | 9600    | 921600  | bps   |
| tCR    | Rise time of data                  |         | 5       | ns    |
| tCF    | Fall time of data                  |         | 5       | ns    |

## 5.2 Slave I2C interface

An I2C compliant interface is available for communication between the RM3100 and BM581 sensors **only**, with an external host CPU. The interface can be operated in slave mode only. It is fully compatible with Fast-Mode-Plus of the I2C industry standard. Since the maximum SCL clock frequency is 1 MHz (in Fast-Mode-Plus), the maximum bit rate is 1 Mbit/s. The interface stretches the clock when slowed down while serving interrupts, therefore the real bit rates may be slightly lower.

Figure 5. I2C chronogram specifications

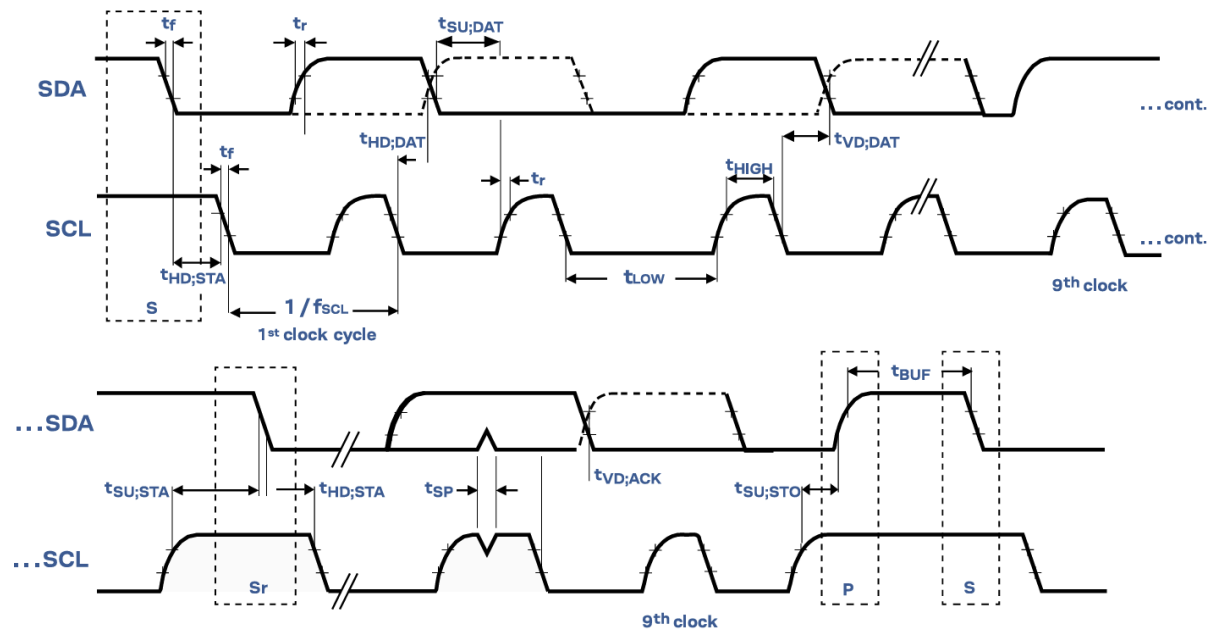
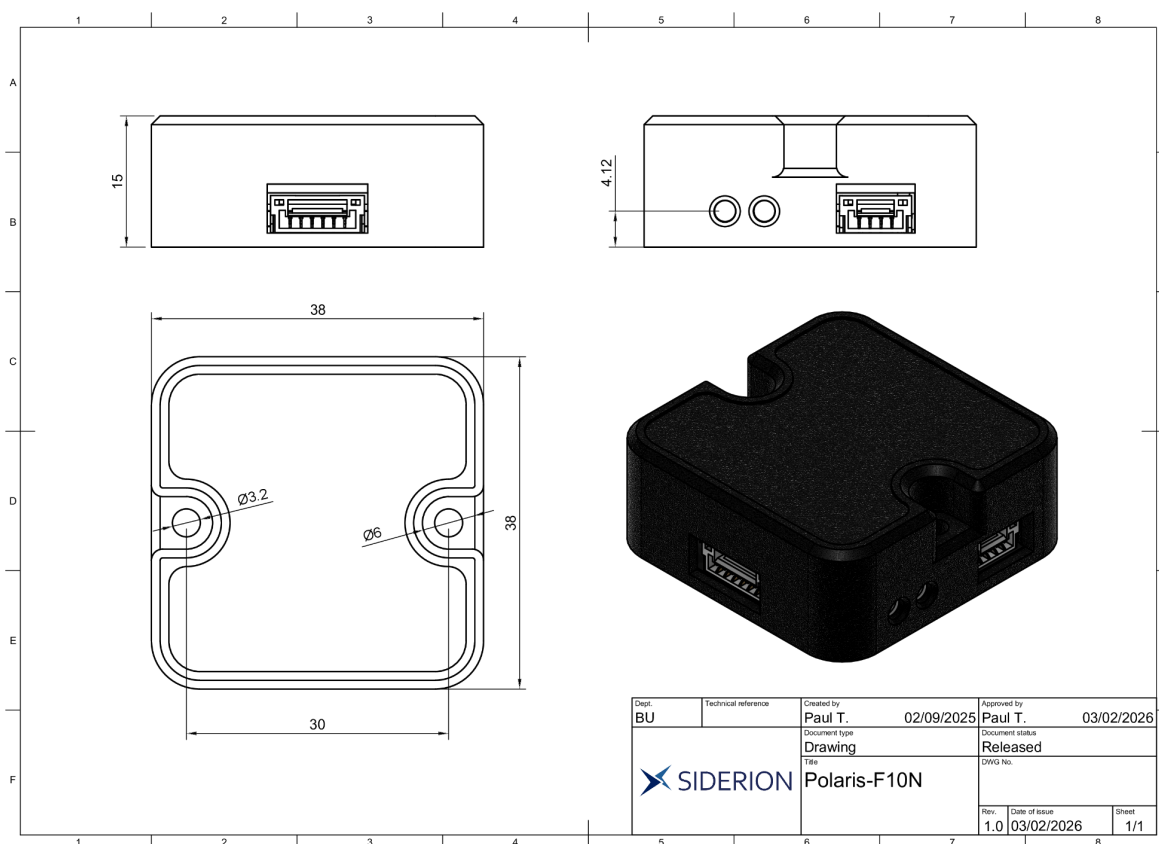


Table 13. Polaris F10N I2C timing specifications

| Symbol | Parameter                | Min     | Max     | Units |
|--------|--------------------------|---------|---------|-------|
| Vil    | LOW-LEVEL input voltage  | Vss-0.3 | 0.3xVdd | V     |
| Vih    | HIGH-LEVEL input voltage | 0.7xVdd | Vdd+0.3 | V     |
| Fscl   | SCL clock frequency      | 0       | 1       | MHz   |

## 6. Mechanical drawings

Figure 6. Polaris F10N v1.0 mechanical drawings



## 7. Revision history

Table 14. Document revision history

| Date        | Revision | Changes                                    |
|-------------|----------|--------------------------------------------|
| 01-Sep-2025 | 1.0      | SiderionDoc_Polaris_F10N / Initial release |
| 26-May-2026 | 1.1      | Figure 1 - Block Diagram added             |

## 8. Appendix

U-blox DAN-F10N datasheet: [DAN-F10N Data sheet](#)

U-blox DAN-F10N integration manual: [DAN-F10N Integration manual](#)

Siderion user's guide:

Siderion tutorials:

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